Advanced Image Processing Methods
Homework 3 Report

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Goal

The task is to investigate how noise reduction and compression parameters affect reconstructed image quality and compression ratio. The pipeline applies Gaussian denoising at multiple kernel sizes, WebP compression at varying quality levels, and evaluates results using full-reference image quality metrics (MSE, PSNR, SSIM).

Method Summary

1. **Dataset Loading (Task 1)**: 10 clean reference images (Kodak dataset, `kodim01`–`kodim10`) and their noisy counterparts are loaded dynamically using `Path.glob`. The clean images are indexed from 1 while the noisy images are indexed from 0, so pairing is done by sorted order rather than by filename index. Each pair is validated for successful loading and matching dimensions.

2. **Noise Reduction (Task 2)**: The noisy images are corrupted with normally-distributed (Gaussian) noise, as indicated by the `norm_dist` suffix in the dataset folder name. A Gaussian low-pass filter (`cv2.GaussianBlur`) is the appropriate choice for this noise type: it optimally attenuates additive Gaussian noise while preserving image structure. Three kernel sizes are evaluated: 3×3, 5×5, and 7×7. Sigma is derived automatically by OpenCV from the kernel size. Every noisy image is denoised with all three kernel sizes.

3. **Image Compression (Task 3)**: Each denoised image is compressed using WebP lossy encoding (`cv2.imencode` with `IMWRITE_WEBP_QUALITY`). The compressed buffer is immediately decoded back (`cv2.imdecode`) to obtain the reconstructed image, simulating a lossy round-trip. Quality levels from 10 to 100 in steps of 10 are evaluated. Every denoised image (3 kernel sizes $\times$ 10 images) is compressed at all 10 quality levels, yielding 300 reconstructed images. The compressed buffer size in bytes is recorded for computing compression ratios.

4. **Quality Measurement (Task 4)**: Each reconstructed image (after denoising + WebP compression) is compared against the corresponding clean reference using three full-reference image quality assessment (FR IQA) metrics:

   - **MSE** – Mean Squared Error, computed as the average of the squared per-pixel differences in float64.
   - **PSNR** – Peak Signal-to-Noise Ratio in dB, via `cv2.PSNR`.
   - **SSIM** – Mean Structural Similarity Index, or MSSIM (Wang et al., 2004). Unlike the global SSIM formula shown in the course slides, this implementation computes a local SSIM at every pixel using an 11×11 Gaussian window ($\sigma = 1.5$), then averages the map. MSSIM is the standard variant used in practice and produces more accurate scores by accounting for spatially varying distortions.

The compression ratio is also recorded as raw_size/compressed_size. All metrics are computed for every (kernel size, quality) combination, yielding $3 \times 10 = 30$ averaged measurement points.

5. **Evaluation (Task 5)**: The averaged metrics (across all 10 images) are plotted as line charts with one line per kernel size and the WebP quality parameter on the x-axis. Four subplots are produced: MSE (↓), PSNR (↑), SSIM (↑), and compression ratio vs. quality. The optimal quality parameter is identified using the Kneedle algorithm (Satopaa et al., 2011), which detects the point of maximum curvature — the "knee" where further increases in quality yield diminishing returns.

Filter Selection Rationale

The Gaussian filter was selected based on two factors:

- **Noise type match**: the dataset folder name (`dobosi_peter_laszlo_norm_dist`) indicates normally-distributed noise. Gaussian blur is the optimal linear filter for additive Gaussian noise, as it minimizes mean squared error in the frequency domain.
- **Simplicity**: among the candidate filters listed in the assignment (averaging, Gaussian, median, bilateral), the Gaussian filter requires only the kernel size parameter, with sigma derived automatically. This makes systematic comparison across kernel sizes straightforward.

A larger kernel provides stronger smoothing and better noise suppression, but at the cost of blurring fine image details. The quality metrics in Task 4 will quantify this trade-off.

Compression Quality Effect

Higher WebP quality values produce larger buffers but preserve more image detail. The table below shows averaged compression ratios (across all 10 images) for each kernel size and selected quality levels:

Quality	3×3	5×5	7×7
10	134:1	153:1	174:1
30	83:1	99:1	118:1
50	59:1	73:1	90:1
70	43:1	55:1	70:1
100	7:1	9:1	13:1

Larger kernels smooth out more detail, which makes the image more compressible (higher ratios). At low quality settings the compressed size is very small but lossy artifacts accumulate on top of the denoising loss. At quality 100, WebP uses near-lossless encoding, resulting in much larger files with minimal additional compression distortion.

Quality Metrics Summary

The following table shows the averaged MSE, PSNR, and SSIM for each kernel size and selected quality levels:

	MSE (↓)			PSNR (↑, dB)			SSIM (↑)		
Quality	3×3	5×5	7×7	3×3	5×5	7×7	3×3	5×5	7×7
10	159	201	246	26.9	26.0	25.2	0.739	0.700	0.662
30	123	169	221	28.0	26.8	25.7	0.788	0.738	0.689
50	106	154	209	28.7	27.2	26.0	0.815	0.760	0.704
70	97	146	201	29.1	27.5	26.2	0.832	0.776	0.716
100	81	129	185	30.0	28.2	26.6	0.857	0.807	0.745

The 3×3 kernel consistently achieves the best quality across all metrics and quality levels, because it removes the least amount of image detail during denoising. However, this comes at the cost of less noise suppression and lower compression ratios compared to larger kernels.

Results

Figure 1 shows the four evaluation plots. All metrics improve monotonically with increasing quality, and the 3×3 kernel consistently achieves the best image quality. Larger kernels yield higher compression ratios because smoother images compress more efficiently, but at the cost of losing fine detail (lower PSNR and SSIM).

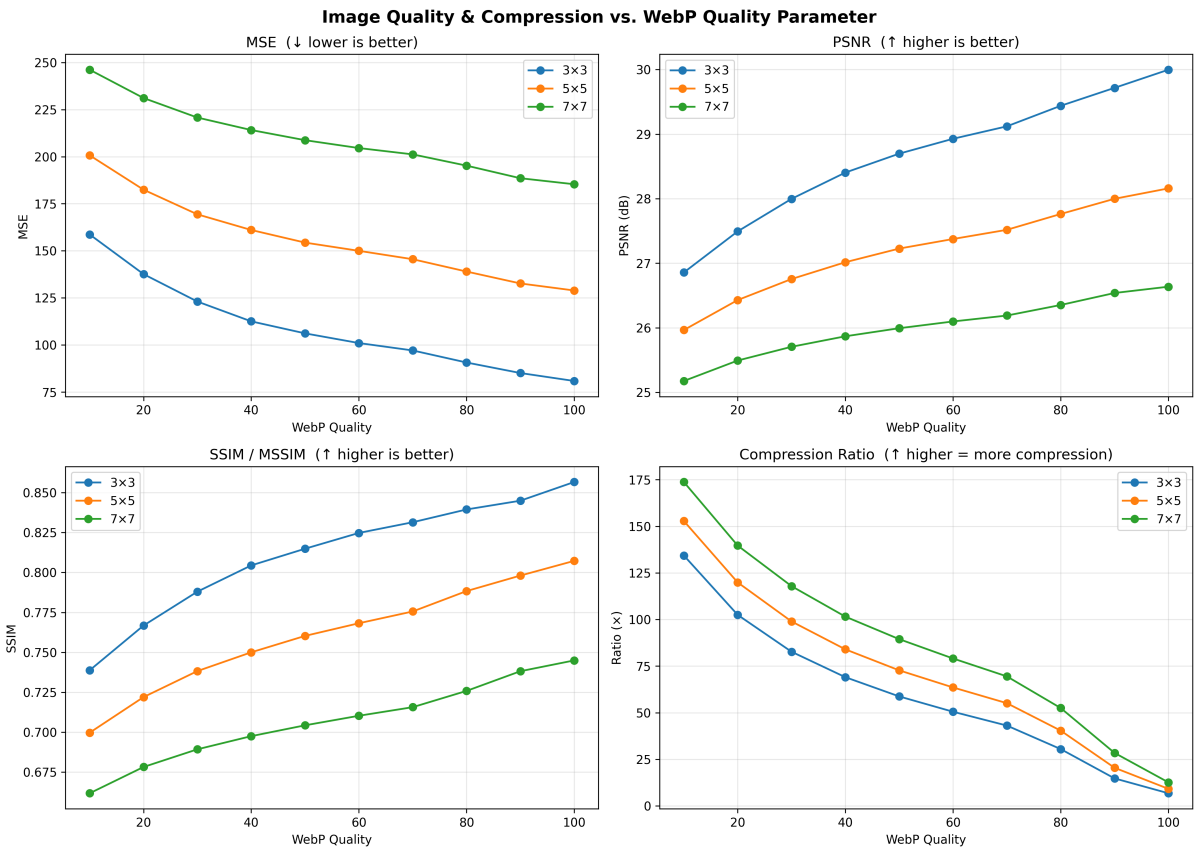

Figure 1: MSE, PSNR, SSIM (MSSIM), and compression ratio averaged over 10 images, for three Gaussian kernel sizes (3×3, 5×5, 7×7).

Optimal Parameter Combination

The assignment asks for the combination providing the best image quality paired with a favourable compression ratio, which leans towards prioritising quality. The 3×3 kernel is the clear winner

on all quality metrics, so the question reduces to choosing the quality parameter.

Identifying the Knee Point

Unlike the example diagrams in the assignment, my curves do not exhibit a single dramatic inflection point. Visual inspection suggests two regions of diminishing returns: one around $q = 30$–40 where the initial steep quality improvement flattens, and another around $q = 70$–80 where quality gains become marginal while file size grows rapidly.

To identify the knee objectively, I applied the Kneedle algorithm (Satopaa et al., 2011), which detects the point of maximum curvature on a normalised curve. For the 3×3 kernel, Kneedle identifies:

- PSNR knee at $q = 40$ (quality gains flatten beyond this point)
- Compression ratio knee at $q = 30$ (compression drops sharply beyond here)

Recommended Combination

Given the assignment's emphasis on quality, I select **kernel 3×3 at WebP quality 40**, achieving:

- PSNR = 28.40 dB
- SSIM (MSSIM) = 0.8044
- Compression ratio = 69.0:1

This sits at the PSNR knee — the point beyond which further quality increases require disproportionately larger files. The second inflection region ($q = 70$–80) could be preferred for use cases that demand higher absolute quality at the expense of compression (e.g. medical imaging or archival), while the $q = 30$–40 region is more suitable when storage efficiency is the primary concern (e.g. web thumbnails). In practice, the ideal compromise depends on the downstream application; the Kneedle-detected knee provides an objective, reproducible starting point for this trade-off.

Reproducibility

From the `homework_3/` directory:

- Install dependencies: `pip install -e .`
- Run pipeline: `python src/mw79on_submission_hw3/main.py`

The script generates output figures in `output/`. This report is provided as `REPORT.pdf`.